KUSHAL DAVE

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EDUCATION

Master of Science in Computer Science **Aug 2024 - May 2026**
The George Washington University, Washington, DC
- GPA: 3.50/4.00
- Relevant Coursework: Machine Learning, Neural Networks & Deep Learning, Natural Language Understanding, Cloud Computing (AWS), Reinforcement Learning, Computer Vision.

Bachelors in Computer Science (AI & ML) **Dec 2020 - Jun 2024**
Gujarat Technological University, Ahmedabad, Gujarat
- GPA: 4.00/4.00

EXPERIENCE

AI/ML Engineer Intern, *ReferU.AI, Arlington Virginia* **Sep 2025 - Present**

• Architected a Generative AI legal research platform using LangChain and PostgreSQL to execute RAG pipelines on proprietary legal datasets.
• Developed autonomous Agentic AI systems via OpenAI Agent Builder to automate client-lawyer matchmaking using Python.
• Engineered scalable document processing and agent orchestration workflows using N8N, Claude Code, and GitHub Copilot.
• Containerized AI features using Docker and managed model deployment versioning via Git and Jenkins.

Technical Product & AI Solutions Intern, *ReferU.AI, Arlington, VA* **May 2025 - Aug 2025**

• Led the lifecycle for a multi-agent GenAI system, defining technical roadmaps for Python and LangChain backend integrations.
• Managed Salesforce data alignment strategies to streamline lead management between AI agents and CRM platforms.
• Directed cross-functional technical pods in Agile/Jira environments to launch 4 features for an Agentic AI marketplace.

Computer Vision Research Engineer Intern, *GTU IDEALAB, Ahmedabad,India* **Jan 2024 - Apr 2024**

• Co-designed a real-time Computer Vision ergonomic assessment system using MoveNet Lightning, TensorFlow, and OpenCV, reducing risk by 20%.
• Optimized Pose Estimation pipelines using OpenCV pre-processing, increasing model accuracy by 15% under occlusion using Keras fine-tuning.
• Secured ₹90,000 SSIP funding and presented research at the IEA 2024 International Congress.

Machine Learning Intern, *RineX.Ai , Ahmedabad, India* **Sept 2023 - Oct 2023**

• Optimized high-volume NLP pipelines (>100k entries), reducing latency by 25% through parallel processing and efficient Python data structures.
• Achieved a 15% accuracy boost in Sentiment Analysis models by implementing TF-IDF vectorization and Scikit-Learn Logistic Regression.

TECHNICAL SKILLS

• AI & GenAI: Generative AI (LLMs), RAG (Retrieval-Augmented Generation), Agentic AI, Prompt Engineering, LangChain, OpenAI Agent Builder, Hugging Face, Reinforcement Learning.
• Computer Vision: OpenCV, YOLOv8, MoveNet Lightning, MediaPipe, Pose Estimation, Object Detection, Image Augmentation, Keras, TensorFlow, PyTorch.
• Data & Databases: PostgreSQL, Vector Databases (Pinecone/Chroma), MySQL, Apache Spark, Data Visualization (Tableau, Power BI).
• Cloud & Infrastructure: AWS (EC2, S3, SageMaker), Google Cloud (GCP), Docker, Kubernetes, N8N (Workflows), Jenkins, CI/CD.
• Languages & CRMs: Python (Advanced), Java, C, SQL, Salesforce Integration, Django, Flask, Git.

Project & Research

• Currency Vision (Deep Learning): Engineered a banknote detection system using YOLOv8m and PyTorch to classify 6 currencies with 95%+ accuracy on 3,000+ custom images.

• Patent: "Method and System for Dynamic Ergonomic Evaluation and Posture Monitoring" (Indian Patent: 202521061742).

• Publication: Sharma, N.K., Dave, K., et al. (2025). "Leveraging MoveNet Lightning for Ergonomic Risk Assessment Through LUBA." Proceedings of IEA 2024, Springer.